

SECTION AAnswer **all** questions.

1. Give the structure of an unsaturated aldehyde of molecular formula C_4H_6O . [1]

2. State a group that will give a positive triiodomethane (iodoform) test and give the observation for a positive result. [2]

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3. (a) 1,2-Diaminoethane reacts as a base.
Explain how this compound acts as a base. [1]

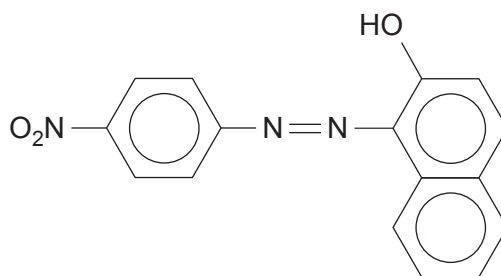
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(b) Give the structure of the organic compound formed when 1 mole of 1,2-diaminoethane reacts with 2 moles of ethanoyl chloride. [1]



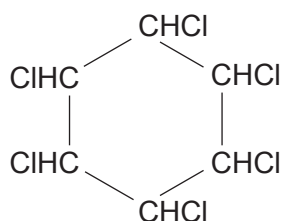
4. The formula of the azo dye, Para Red, is shown below.



Give the structure of the compound that couples with 4-nitrobenzenediazonium chloride to give this dye.

[1]

5. Hexachlorocyclohexane can be used as an insecticide.

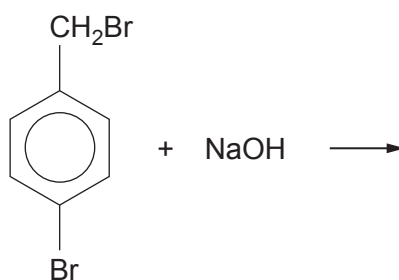


Deduce the empirical formula of this compound.

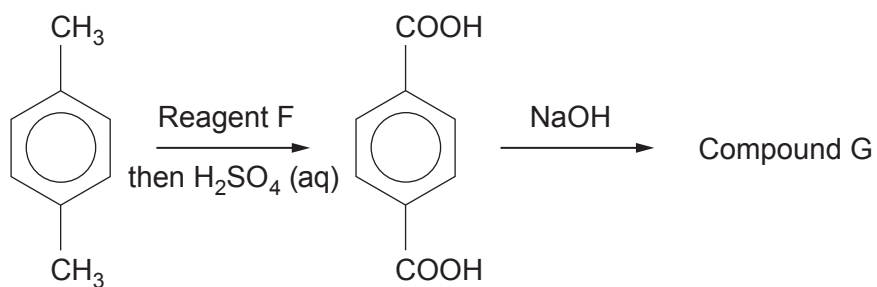
[1]

6. Give the structure of the organic product of the reaction below.

[1]



7. Study the reaction sequence below and answer the **two** questions that follow.



(a) Reagent F is

[1]

(b) Give the structure of Compound G.

[1]

